

Antioxidant diet supplementation enhances aerobic performance in amateur sportsmen.

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The effects of antioxidant diet supplements on blood lactate concentration and on the aerobic and anaerobic thresholds and their adaptations to training were analysed. Fifteen amateur male athletes were randomly assigned to either a placebo group or an antioxidant-supplemented group (90 days supplementation with 500 mg x day⁻¹ of vitamin E and 30 mg x day⁻¹ of beta-carotene, and the last 15 days also with 1 g x day⁻¹ of vitamin C). Before and after the antioxidant supplements, the sportsmen performed a maximal exercise test on a cycle ergometer and maximal and submaximal physiological parameters were assessed together with blood lactate concentration. Maximal oxygen uptake (VO₂max), maximal blood lactate concentration, and the maximal workload attained rose significantly in both groups after the 3 months of training. At the end of the study, maximal blood lactate concentration was lower in the group that took supplements than in the placebo group. The percentage of VO₂max attained at the anaerobic threshold rose significantly in both groups after 3 months of training, although the final value in the supplemented group was higher than that in the placebo group. Antioxidant diet supplements induced lower increases in blood lactate concentration after a maximal exercise test and could improve the efficiency in which aerobic energy is obtained.